

## VIOLET GROVE, AB, Canada

WMO: 710620

Lat: 53.144N

Lon: 115.126W

Elev: 2963

StdP: 13.19

Time Zone: -7.00 (NAM)

Period: 01-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |       | Humidification DP/MCDB and HR |     |       |       |     |       | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-------|-------------------------------|-----|-------|-------|-----|-------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |       | 99.6%                         |     |       | 99%   |     |       | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%   | DP                            | HR  | MCDB  | DP    | HR  | MCDB  | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)   | (d)                           | (e) | (f)   | (g)   | (h) | (i)   | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 12            | -19.7      | -14.0 | -26.4                         | 1.4 | -19.1 | -20.6 | 2.0 | -13.4 | 23.2                  | 30.4 | 20.2 | 28.2 | 4.7                   | 250  | 0.642 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 21.8                   | 81.0            | 62.4 | 77.9 | 60.7 | 75.0 | 59.4 | 65.0                | 77.3 | 62.8 | 74.8 | 60.9 | 72.3 | 6.7                  | 120  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 60.2                            | 86.8 | 71.4 | 57.9 | 79.8 | 68.1 | 56.0 | 74.6 | 65.8 | 31.6          | 77.4 | 29.9 | 75.0 | 28.4 | 72.4 | 74.5              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |    |    |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|----|----|
|                   |      |      | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |    |    |
| 1%                | 2.5% | 5%   | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  | DB | WB |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |    |    |
| (4) 20.1          | 16.9 | 14.3 | -27.3                      | 87.4 | 5.8                | 3.2 | -31.5                                              | 89.7 | -34.9      | 91.5 | -38.2      | 93.3 | -42.4      | 95.6 |    |    |
| (5)               |      |      | -27.6                      | 68.1 | 5.7                | 2.0 | -31.7                                              | 69.5 | -35.1      | 70.7 | -38.3      | 71.8 | -42.5      | 73.3 |    |    |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |  |  |
|------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|--|--|
|                                                                        |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |  |  |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 39.1   | 18.3  | 19.3  | 27.0  | 39.3  | 49.6  | 56.5  | 61.4  | 59.4  | 51.1  | 40.0  | 28.0  | 18.0  |  |  |
|                                                                        | DBStd       | 19.10  | 15.79 | 14.54 | 13.89 | 9.12  | 7.94  | 5.53  | 5.41  | 5.86  | 7.96  | 8.98  | 12.19 | 13.71 |  |  |
|                                                                        | HDD50       | 5069   | 982   | 860   | 712   | 334   | 105   | 8     | 1     | 5     | 80    | 327   | 662   | 993   |  |  |
|                                                                        | HDD65       | 9502   | 1447  | 1280  | 1177  | 772   | 477   | 260   | 136   | 189   | 419   | 776   | 1111  | 1458  |  |  |
|                                                                        | CDD50       | 1091   | 0     | 0     | 1     | 12    | 94    | 204   | 356   | 295   | 112   | 16    | 1     | 0     |  |  |
|                                                                        | CDD65       | 49     | 0     | 0     | 0     | 0     | 1     | 6     | 26    | 14    | 2     | 0     | 0     | 0     |  |  |
|                                                                        | CDH74       | 834    | 0     | 0     | 0     | 3     | 46    | 98    | 376   | 222   | 83    | 6     | 0     | 0     |  |  |
|                                                                        | CDH80       | 135    | 0     | 0     | 0     | 0     | 4     | 11    | 77    | 33    | 10    | 0     | 0     | 0     |  |  |
|                                                                        | WSAvg       | 6.7    | 6.6   | 6.5   | 6.9   | 7.7   | 7.8   | 6.9   | 6.4   | 5.9   | 6.1   | 6.7   | 6.4   | 6.5   |  |  |
|                                                                        | PrecAvg     | 21.6   | 1.0   | 0.7   | 0.7   | 1.3   | 2.5   | 3.7   | 4.2   | 2.8   | 1.7   | 1.1   | 0.9   | 0.6   |  |  |
| (15) Precipitation                                                     | PrecMax     | 24.1   | 3.1   | 1.5   | 1.7   | 2.8   | 4.0   | 5.9   | 9.8   | 5.1   | 3.7   | 2.8   | 2.4   | 1.4   |  |  |
|                                                                        | PrecMin     | 17.6   | 0.1   | 0.1   | 0.1   | 0.3   | 1.1   | 1.1   | 1.8   | 1.1   | 0.4   | 0.3   | 0.2   | 0.2   |  |  |
|                                                                        | PrecStd     | 1.9    | 0.8   | 0.4   | 0.4   | 0.6   | 0.7   | 1.3   | 1.8   | 0.9   | 0.9   | 0.7   | 0.7   | 0.4   |  |  |
|                                                                        |             |        |       |       |       |       |       |       |       |       |       |       |       |       |  |  |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB     | 51.5  | 52.3  | 59.1  | 71.8  | 79.2  | 81.3  | 86.6  | 84.2  | 81.1  | 72.0  | 58.9  | 48.6  |  |  |
|                                                                        |             | MCWB   | 39.6  | 38.9  | 42.8  | 50.5  | 55.7  | 62.0  | 65.4  | 64.1  | 60.2  | 52.4  | 44.7  | 38.0  |  |  |
|                                                                        | 2%          | DB     | 46.0  | 46.0  | 52.1  | 64.8  | 74.0  | 76.7  | 81.9  | 79.5  | 76.2  | 63.1  | 52.0  | 43.8  |  |  |
|                                                                        |             | MCWB   | 36.8  | 35.5  | 39.1  | 47.1  | 54.0  | 59.0  | 64.3  | 62.4  | 58.5  | 48.8  | 41.1  | 35.2  |  |  |
|                                                                        | 5%          | DB     | 42.1  | 42.0  | 47.6  | 59.3  | 69.9  | 73.1  | 78.2  | 76.1  | 71.0  | 58.3  | 46.8  | 39.6  |  |  |
|                                                                        |             | MCWB   | 34.4  | 33.4  | 36.6  | 43.9  | 51.6  | 57.2  | 62.4  | 61.1  | 56.2  | 46.0  | 37.7  | 32.8  |  |  |
|                                                                        | 10%         | DB     | 38.1  | 38.0  | 43.8  | 54.6  | 65.6  | 69.5  | 74.7  | 72.6  | 65.2  | 53.6  | 42.1  | 35.7  |  |  |
|                                                                        |             | MCWB   | 32.1  | 31.1  | 34.7  | 41.4  | 49.8  | 55.5  | 60.7  | 59.9  | 53.1  | 43.3  | 35.1  | 29.9  |  |  |
|                                                                        | 0.4%        | WB     | 40.1  | 39.3  | 43.3  | 51.2  | 57.8  | 63.8  | 68.7  | 66.7  | 62.4  | 53.9  | 45.1  | 38.5  |  |  |
|                                                                        |             | MCDB   | 50.2  | 50.7  | 57.4  | 70.4  | 74.7  | 77.4  | 80.4  | 78.5  | 79.1  | 70.4  | 57.4  | 47.4  |  |  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 2%          | WB     | 37.3  | 36.3  | 39.6  | 48.0  | 54.9  | 60.9  | 65.9  | 64.4  | 59.1  | 49.5  | 41.5  | 35.7  |  |  |
|                                                                        |             | MCDB   | 44.9  | 45.1  | 50.8  | 63.4  | 70.8  | 73.2  | 78.4  | 76.6  | 73.2  | 61.9  | 50.7  | 43.0  |  |  |
|                                                                        | 5%          | WB     | 34.9  | 33.7  | 37.2  | 44.7  | 52.9  | 58.8  | 63.6  | 62.2  | 56.7  | 46.7  | 38.6  | 33.0  |  |  |
|                                                                        |             | MCDB   | 41.7  | 41.7  | 46.4  | 57.9  | 67.2  | 70.3  | 75.5  | 73.6  | 69.4  | 56.7  | 45.6  | 39.1  |  |  |
|                                                                        | 10%         | WB     | 32.3  | 31.3  | 35.1  | 42.0  | 50.8  | 56.7  | 61.6  | 60.2  | 54.2  | 44.0  | 35.7  | 30.4  |  |  |
|                                                                        |             | MCDB   | 37.7  | 37.4  | 43.2  | 52.8  | 63.8  | 67.2  | 72.9  | 70.9  | 64.0  | 52.2  | 41.5  | 35.4  |  |  |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR   | 17.6  | 20.1  | 20.1  | 20.6  | 22.2  | 20.7  | 21.8  | 21.7  | 21.3  | 18.7  | 15.8  | 16.2  |  |  |
|                                                                        |             | MCDBR  | 19.2  | 23.0  | 24.8  | 29.0  | 29.8  | 26.5  | 27.9  | 27.6  | 31.1  | 26.9  | 20.8  | 18.3  |  |  |
|                                                                        |             | MCWBR  | 12.6  | 14.8  | 14.8  | 15.4  | 14.4  | 12.8  | 13.0  | 13.5  | 15.6  | 15.0  | 12.5  | 12.5  |  |  |
|                                                                        |             | MCDBR  | 19.0  | 22.4  | 23.4  | 27.7  | 26.8  | 23.4  | 25.8  | 25.7  | 29.0  | 25.8  | 19.9  | 18.0  |  |  |
|                                                                        | 5% WB       | MCWBR  | 13.3  | 14.6  | 14.4  | 15.0  | 13.6  | 12.0  | 13.0  | 13.5  | 14.9  | 14.7  | 12.6  | 12.9  |  |  |
|                                                                        |             |        |       |       |       |       |       |       |       |       |       |       |       |       |  |  |
| (40) Clear-Sky Solar Irradiance                                        | taub        |        | 0.246 | 0.256 | 0.285 | 0.326 | 0.338 | 0.350 | 0.348 | 0.355 | 0.312 | 0.280 | 0.253 | 0.236 |  |  |
|                                                                        | taud        |        | 2.427 | 2.477 | 2.471 | 2.405 | 2.417 | 2.397 | 2.454 | 2.394 | 2.515 | 2.592 | 2.518 | 2.396 |  |  |
|                                                                        | Ebn at Noon |        | 246   | 280   | 291   | 288   | 289   | 285   | 284   | 275   | 277   | 267   | 244   | 227   |  |  |
|                                                                        | Edh at Noon |        | 16    | 22    | 27    | 33    | 34    | 35    | 33    | 33    | 26    | 19    | 15    | 14    |  |  |
| (44) All-Sky Solar Radiation                                           | RadAvg      |        | 308   | 621   | 1079  | 1464  | 1747  | 1859  | 1924  | 1596  | 1111  | 649   | 330   | 228   |  |  |
|                                                                        | RadStd      |        | 22    | 45    | 66    | 128   | 125   | 130   | 118   | 107   | 116   | 68    | 29    | 18    |  |  |

## Historical Trends

|      |                         | DBAvg  |        | Heating |       | Cooling |       | Degree-Days |       |       |     |
|------|-------------------------|--------|--------|---------|-------|---------|-------|-------------|-------|-------|-----|
|      |                         | 99% DB | 99% DP | 1% DB   | 1% WB | 1% DP   | HDD50 | HDD65       | CDD50 | CDD65 |     |
| (46) | Station Only            | N/A    | N/A    | N/A     | N/A   | N/A     | N/A   | N/A         | N/A   | N/A   | -34 |
| (47) | Regional (47 neighbors) | N/A    | N/A    | N/A     | N/A   | N/A     | N/A   | N/A         | N/A   | N/A   | N/A |

Nomenclature: See separate page